

Generative AI: Assignment 1

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This notebook is built with **LangChain** + a **local Ollama model** (`llama3.2`) so it can run completely offline/free and without hitting Groq rate limits, per the assignment's note.

- **Part 1** — Topic Detection & Summarization of BBC News Articles (45 marks)
- **Part 2** — Job Postings Analysis: Role Categorization & Requirements Extraction (55 marks)
- **Bonus** — Full-dataset runs for both parts (20 marks), included at the end, checkpointed so they can resume if interrupted.

Setup

```
In [1]: import json
import re
import time
from pathlib import Path

import pandas as pd
from tqdm.auto import tqdm

from langchain_core.prompts import ChatPromptTemplate, FewShotChatMessagePromptTemplate
from langchain_core.output_parsers import StrOutputParser
from langchain_ollama import ChatOllama
from pydantic import BaseModel, Field

pd.set_option("display.max_colwidth", 120)
```

```
In [2]: # Ollama runs locally; OLLAMA_HOST is set to 0.0.0.0 (server bind address) in this environment,
# which is NOT a valid address for the *client* to connect to, so we point the client at
# localhost explicitly.
OLLAMA_BASE_URL = "http://localhost:11434"
MODEL_NAME = "llama3.2"

llm = ChatOllama(model=MODEL_NAME, base_url=OLLAMA_BASE_URL, temperature=0)
```

```
# num_predict caps output length: in JSON mode llama3.2 occasionally loops emitting output
# until the context fills (~40k tokens, 10+ minutes per call), stalling the whole run.
llm_json = ChatOllama(model=MODEL_NAME, base_url=OLLAMA_BASE_URL, temperature=0, format="json", num_predict=512)

# Quick smoke test
print(llm.invoke("Reply with just the word: ready").content)
```

ready

Part 1: Topic Detection and Summarization of News Articles (45 marks)

Step 1: Load the Dataset

```
In [3]: bbc_df_full = pd.read_csv("bbc-news-data.csv", sep="\t")
print("Full dataset shape:", bbc_df_full.shape)
print(bbc_df_full["category"].value_counts())

bbc_df = bbc_df_full.head(30).reset_index(drop=True)
bbc_df.insert(0, "Article_ID", bbc_df.index)
bbc_df.head()
```

Full dataset shape: (2225, 4)

category

sport 511

business 510

politics 417

tech 401

entertainment 386

Name: count, dtype: int64

Out[3]:

	Article_ID	category	filename	title	content
0	0	business	001.txt	Ad sales boost Time Warner profit	Quarterly profits at US media giant TimeWarner jumped 76% to \$1.13bn (£600m) for the three months to December, from...
1	1	business	002.txt	Dollar gains on Greenspan speech	The dollar has hit its highest level against the euro in almost three months after the Federal Reserve head said th...
2	2	business	003.txt	Yukos unit buyer faces loan claim	The owners of embattled Russian oil giant Yukos are to ask the buyer of its former production unit to pay back a \$9...
3	3	business	004.txt	High fuel prices hit BA's profits	British Airways has blamed high fuel prices for a 40% drop in profits. Reporting its results for the three months ...
4	4	business	005.txt	Pernod takeover talk lifts Domecq	Shares in UK drinks and food firm Allied Domecq have risen on speculation that it could be the target of a takeover...

Step 2: Define the Topic Classification Task (10 marks)

We use a `ChatPromptTemplate` with short category definitions and one few-shot example per category (plus an economy-vs-politics edge case), then a simple LCEL chain (`prompt | llm | StrOutputParser()`) so the model returns a single category label.

```
In [4]: TOPIC_CATEGORIES = ["Business", "Entertainment", "Politics", "Sport", "Tech"]

# One example per category, plus an economy-vs-politics edge case: without these, Llama3.2
# Labelled many business articles as Politics/Tech and never predicted Sport or Entertainment
# (accuracy on the first 30 articles rose from 67% to 87% with this prompt).
topic_few_shot_examples = [
    {"article": "Shares in the carmaker fell 5% after it reported lower quarterly profits and "
               "warned that sales would slow next year.", "category": "Business"},
    {"article": "The central bank raised interest rates for the third time this year as inflation "
               "and consumer spending kept rising.", "category": "Business"},
    {"article": "The actor won best actress at the film awards for her role in the period drama, "
               "which also took the best picture prize.", "category": "Entertainment"},
    {"article": "The prime minister faced MPs' questions over the proposed election date as "
               "opposition parties attacked the government's plans.", "category": "Politics"},
    {"article": "The striker scored a hat-trick as the home team thrashed their rivals 4-0 in "
               "front of a record crowd at the stadium.", "category": "Sport"},

```

```

        {"article": "The firm unveiled a new smartphone with a faster chip and said online downloads "
            "of its music software had doubled.", "category": "Tech"},
    ]

topic_example_prompt = ChatPromptTemplate.from_messages(
    [("human", "Article: {article}"), ("ai", "{category}")]
)
topic_few_shot_prompt = FewShotChatMessagePromptTemplate(
    example_prompt=topic_example_prompt,
    examples=topic_few_shot_examples,
)

topic_prompt = ChatPromptTemplate.from_messages(
    [
        ("system",
            "You are a news topic classifier. Analyze the following news article and identify its topic as "
            "ONE of the following categories: Business, Entertainment, Politics, Sport, or Tech.\n"
            "- Business: companies, earnings, markets, the economy, trade, jobs, prices, interest rates, finance.\n"
            "- Entertainment: film, music, TV, celebrities, books, theatre, arts and awards.\n"
            "- Politics: government, elections, parliament, political parties, ministers and legislation.\n"
            "- Sport: athletes, matches, tournaments, clubs and sporting events.\n"
            "- Tech: technology, internet, software, gadgets, video games, telecoms and science.\n"
            "Economic news is Business even when governments or central banks are mentioned; only choose "
            "Politics when the article is mainly about politicians, elections or law-making.\n"
            "Reply with ONLY the single category label, nothing else."),
        topic_few_shot_prompt,
        ("human", "Article: {article}"),
    ]
)

topic_chain = topic_prompt | llm | StrOutputParser()

def classify_topic(article_text: str) -> str:
    raw = topic_chain.invoke({"article": article_text[:3000]}).strip()
    # Normalize to one of the known labels in case the model adds extra words/punctuation.
    for cat in TOPIC_CATEGORIES:
        if cat.lower() in raw.lower():
            return cat
    return raw.split()[0] if raw else "Unknown"

```

```
# Expected Output: show it works for a sample datapoint.
sample = bbc_df.iloc[0]
sample_topic = classify_topic(sample["content"])
print("Actual category:", sample["category"])
print("Predicted topic :", sample_topic)
```

Actual category: business
Predicted topic : Business

Step 3: Define the Summarization Task (10 marks)

```
In [5]: summary_prompt = ChatPromptTemplate.from_messages(
    [
        ("system",
         "You summarize news articles factually and concisely, capturing who/what/when/where/why "
         "as applicable, without personal commentary or opinions. "
         "Respond with only the summary itself, with no introduction such as 'Here is a summary'. "),
        ("human",
         "Summarize the main points of the following news article in 2-3 sentences.\n\n"
         "Article:\n{article}"),
    ]
)
summary_chain = summary_prompt | llm | StrOutputParser()

# Small models often prepend a lead-in such as "Here is a summary of the article in 3 sentences:";
# strip it so the Summary column holds only the summary itself.
SUMMARY_LEAD_IN = re.compile(r"^\s*(?:here(?:'s| is| are)|below is)[^\n]*?:\s*", re.IGNORECASE)

def summarize_article(article_text: str) -> str:
    summary = summary_chain.invoke({"article": article_text[:4000]}).strip()
    return SUMMARY_LEAD_IN.sub("", summary).strip()

# Expected Output: show it works for a sample datapoint.
sample_summary = summarize_article(sample["content"])
print("Summary:\n", sample_summary)
```

Summary:

Time Warner's quarterly profits rose 76% to \$1.13 billion, driven by sales of high-speed internet connections and higher advertisement sales, with the company also benefiting from one-off gains. The firm's internet business, AOL, saw a 2% increase in sales but lost 464,000 subscribers in the fourth quarter. Time Warner now owns 8% of Google and is projecting 5% operating earnings growth for 2005.

Step 4: Key Entity Extraction (10 marks)

We ask the model to return **structured JSON** (via Ollama's `format="json"` mode) so the `Key_Entities` column can hold a clean Python list rather than free-form text.

```
In [6]: class EntityExtraction(BaseModel):
        entities: list[str] = Field(
            description="Important people, organizations, and places mentioned in the article."
        )

        entity_prompt = ChatPromptTemplate.from_messages(
            [
                ("system",
                 "From the article, list the names of any important people, organizations, or places "
                 "mentioned. Respond with ONLY a JSON object of the form "
                 "'{"entities": ["Name1", "Name2", ...]}'. If none are found, return an empty list."),
                ("human", "Article:\n{article}"),
            ]
        )
        entity_chain = entity_prompt | llm_json | StrOutputParser()

        def extract_entities(article_text: str) -> list[str]:
            raw = entity_chain.invoke({"article": article_text[:4000]}).strip()
            try:
                data = json.loads(raw)
                entities = data.get("entities", [])
                return [str(e).strip() for e in entities if str(e).strip()]
            except (json.JSONDecodeError, AttributeError):
                # Fallback: split on commas/newlines if the model didn't return clean JSON.
                return [e.strip("- \t") for e in re.split(r"[,\n]", raw) if e.strip("- \t")]
```

```
# Expected Output: show it works for a sample datapoint.
sample_entities = extract_entities(sample["content"])
print("Key entities:", sample_entities)
```

Key entities: ['Richard Parsons', 'TimeWarner', 'Google', 'Warner Bros', 'AOL', 'US Securities Exchange Commission', 'Bertelsman n', 'Lord of the Rings', 'Alexander', 'Catwoman', 'SEC']

Step 5: Update the DataFrame with Results (15 marks)

Apply classification, summarization, and entity extraction to every row of the 30-article DataFrame, with a checkpoint file so re-running the notebook doesn't redo completed rows.

```
In [7]: BBC_CHECKPOINT = Path("bbc_news_first30_checkpoint.csv")

def run_bbc_pipeline(df: pd.DataFrame, checkpoint_path: Path, desc: str) -> pd.DataFrame:
    if checkpoint_path.exists():
        result = pd.read_csv(checkpoint_path)
    else:
        result = df.copy()
        result["Detected_Topic"] = pd.NA
        result["Summary"] = pd.NA
        result["Key_Entities"] = pd.NA

    for idx in tqdm(result.index, desc=desc):
        if pd.notna(result.loc[idx, "Detected_Topic"]):
            continue # already processed (resume support)
        text = result.loc[idx, "content"]
        try:
            topic = classify_topic(text)
            summ = summarize_article(text)
            ents = extract_entities(text)
        except Exception as e: # keep going even if Ollama hiccups on one row
            print(f"Row {idx} failed: {e}")
            topic, summ, ents = "Error", "Error", []
        result.loc[idx, "Detected_Topic"] = topic
        result.loc[idx, "Summary"] = summ
        result.loc[idx, "Key_Entities"] = json.dumps(ents)
        result.to_csv(checkpoint_path, index=False) # checkpoint after every row

result["Key_Entities"] = result["Key_Entities"].apply(json.loads)
```

```
return result
```

```
bbc_results = run_bbc_pipeline(bbc_df, BBC_CHECKPOINT, "Part 1: first 30 articles")  
bbc_results.head()
```

Out[7]:

	Article_ID	category	filename	title	content	Detected_Topic	Summary	Key_Entities
0	0	business	001.txt	Ad sales boost Time Warner profit	Quarterly profits at US media giant TimeWarner jumped 76% to \$1.13bn (£600m) for the three months to December, from...	Business	Time Warner's quarterly profits rose 76% to \$1.13 billion, driven by sales of high-speed internet connections and hi...	[Richard Parsons, TimeWarner, Google, Warner Bros, AOL, US Securities Exchange Commission, Bertelsmann, Lord of the ...
1	1	business	002.txt	Dollar gains on Greenspan speech	The dollar has hit its highest level against the euro in almost three months after the Federal Reserve head said th...	Business	The US dollar has reached its highest level against the euro in almost three months after Federal Reserve Chairman A...	[Alan Greenspan, Robert Sinche, Federal Reserve, Bank of America, G7, China, White House]
2	2	business	003.txt	Yukos unit buyer faces loan claim	The owners of embattled Russian oil giant Yukos are to ask the buyer of its former production unit to pay back a \$9...	Business	The owners of Yukos, Menatep Group, plan to ask the buyer of its former production unit, Rosneft, to repay a \$900m l...	[Jamie Firestone, Tim Osborne, Mikhail Khodorkovsky, Rosneft, Yukos, Menatep Group, Reuters, US, Russia, Moscow, Yug...
3	3	business	004.txt	High fuel prices hit BA's profits	British Airways has blamed high fuel prices for a 40% drop in profits. Reporting its results for the three months ...	Business	British Airways reported a 40% drop in profits to £75m for the three months to December 2004, blaming high fuel pric...	[Rod Eddington, Mike Powell, Martin Broughton, Nick Van den Brul]
4	4	business	005.txt	Pernod takeover talk lifts Domecq	Shares in UK drinks and food firm Allied Domecq have risen on speculation that it could be the target of a takeover...	Business	Shares in UK drinks and food firm Allied Domecq rose 4% on speculation that France's Pernod Ricard is considering a ...	[Pernod Ricard, Wall Street Journal, Financial Times, Diageo, Seagram, LVMH, Glenmorangie, Jacob's Creek, Chivas Reg...

In [8]: # Final merged dataframe: original + new columns together.

```
bbc_final = bbc_results[  
    ["Article_ID", "category", "filename", "title", "content", "Detected_Topic", "Summary", "Key_Entities"]  
]  
bbc_final.to_csv("bbc_news_analyzed_first30.csv", index=False)
```

```
print(bbc_final.shape)
bbc_final.head(10)
```

(30, 8)

Out[8]:

	Article_ID	category	filename	title	content	Detected_Topic	Summary	Key_Entities
0	0	business	001.txt	Ad sales boost Time Warner profit	Quarterly profits at US media giant TimeWarner jumped 76% to \$1.13bn (£600m) for the three months to December, from...	Business	Time Warner's quarterly profits rose 76% to \$1.13 billion, driven by sales of high-speed internet connections and hi...	[Richard Parsons, TimeWarner, Google, Warner Bros, AOL, US Securities Exchange Commission, Bertelsmann, Lord of the ...
1	1	business	002.txt	Dollar gains on Greenspan speech	The dollar has hit its highest level against the euro in almost three months after the Federal Reserve head said th...	Business	The US dollar has reached its highest level against the euro in almost three months after Federal Reserve Chairman A...	[Alan Greenspan, Robert Sinche, Federal Reserve, Bank of America, G7, China, White House]
2	2	business	003.txt	Yukos unit buyer faces loan claim	The owners of embattled Russian oil giant Yukos are to ask the buyer of its former production unit to pay back a \$9...	Business	The owners of Yukos, Menatep Group, plan to ask the buyer of its former production unit, Rosneft, to repay a \$900m l...	[Jamie Firestone, Tim Osborne, Mikhail Khodorkovsky, Rosneft, Yukos, Menatep Group, Reuters, US, Russia, Moscow, Yug...
3	3	business	004.txt	High fuel prices hit BA's profits	British Airways has blamed high fuel prices for a 40% drop in profits. Reporting its results for the three months ...	Business	British Airways reported a 40% drop in profits to £75m for the three months to December 2004, blaming high fuel pric...	[Rod Eddington, Mike Powell, Martin Broughton, Nick Van den Brul]
4	4	business	005.txt	Pernod takeover talk lifts Domecq	Shares in UK drinks and food firm Allied Domecq have risen on speculation that it could be the target of a takeover...	Business	Shares in UK drinks and food firm Allied Domecq rose 4% on speculation that France's Pernod Ricard is considering a ...	[Pernod Ricard, Wall Street Journal, Financial Times, Diageo, Seagram, LVMH, Glenmorangie, Jacob's Creek, Chivas Reg...
5	5	business	006.txt	Japan narrowly escapes recession	Japan's economy teetered on the brink of a technical recession in the three months to September, figures show. Rev...	Business	Japan's economy experienced a technical recession in the three months to September, with revised figures showing 0.1...	[Heizo Takenaka, Paul Sheard, Lehman Brothers, Japan]
6	6	business	007.txt	Jobs growth still slow in the US	The US created fewer jobs than expected in January, but a fall in jobseekers pushed the unemployment rate to its lo...	Business	The US economy added 146,000 jobs in January, below market expectations of 190,000, but the unemployment rate fell t...	[Herbert Hoover, Rick Egelton, Ken Mayland, BMO Financial Group, ClearView Economics, President Bush, Labor Departme...

	Article_ID	category	filename	title	content	Detected_Topic	Summary	Key_Entities
7	7	business	008.txt	India calls for fair trade rules	India, which attends the G7 meeting of seven leading industrialised nations on Friday, is unlikely to be cowed by i...	Politics	India's finance minister, Palaniappan Chidambaram, criticized the restrictive trade policies of the G7 nations durin...	[Palaniappan Chidambaram, Gordon Brown, United Nations, World Bank, IMF, G7, G20, China, Brazil, South Africa, Russi...
8	8	business	009.txt	Ethiopia's crop production up 24%	Ethiopia produced 14.27 million tonnes of crops in 2004, 24% higher than in 2003 and 21% more than the average of t...	Business	Ethiopia's crop production increased by 21% in 2004, reaching 14.27 million tonnes, with good rains, fertilizer use,...	[Henri Josserand, Food and Agriculture Organisation, World Food Programme, Ethiopia]
9	9	business	010.txt	Court rejects \$280bn tobacco case	A US government claim accusing the country's biggest tobacco companies of covering up the effects of smoking has be...	Politics	A US appeals court has rejected a \$280bn claim against the country's biggest tobacco companies, finding that the cas...	[Altria Group, RJ Reynolds Tobacco, Lorillard Tobacco, Liggett Group, Brown and Williamson, Clinton administration, ...]

```
In [9]: # Quick accuracy check against the dataset's ground-truth category labels.
accuracy = (bbc_final["category"].str.lower() == bbc_final["Detected_Topic"].str.lower()).mean()
print(f"Topic classification accuracy on first 30 articles: {accuracy:.1%}")
```

Topic classification accuracy on first 30 articles: 86.7%

Part 2: Job Postings Analysis – Role Categorization and Requirements Extraction (55 marks)

Step 1: Load the Dataset

```
In [10]: jobs_df_full = pd.read_csv("job_title_des.csv")
jobs_df_full = jobs_df_full.rename(
    columns={"Job Title": "Job_Title", "Job Description": "Job_Description"}
)
jobs_df_full = jobs_df_full.drop(columns=[c for c in jobs_df_full.columns if c.startswith("Unnamed")])
print("Full dataset shape:", jobs_df_full.shape)
```

```
jobs_df = jobs_df_full.head(25).reset_index(drop=True)
jobs_df.head()
```

Full dataset shape: (2277, 2)

Out[10]:

	Job_Title	Job_Description
0	Flutter Developer	We are looking for hire experts flutter developer. So you are eligible this post then apply your resume.\nJob Types:...
1	Django Developer	PYTHON/DJANGO (Developer/Lead) - Job Code(PDJ - 04)\nStrong Python experience in API development (REST/RPC).\nExperi...
2	Machine Learning	Data Scientist (Contractor)\n\nBangalore, IN\n\nResponsibilities\n\nWe are looking for a capable data scientist to j...
3	iOS Developer	JOB DESCRIPTION:\n\nStrong framework outside of iOS is always a plus\n\niOS experience and generalist engineers with...
4	Full Stack Developer	job responsibility full stack engineer – react role make impact petsmart transforming engineering team meet need rap...

Step 2: Define the Job Category Classification Task (10 marks)

```
In [11]: JOB_CATEGORIES = [
    "Technology/IT", "Finance", "Marketing", "Healthcare", "Sales",
    "Human Resources", "Operations", "Design", "Education", "Other",
]

job_category_few_shot = [
    {
        "title": "Senior Data Analyst",
        "description": "Analyze sales data, build dashboards, SQL and Python required.",
        "category": "Technology/IT",
    },
    {
        "title": "Registered Nurse",
        "description": "Provide patient care in a hospital ward, monitor vitals.",
        "category": "Healthcare",
    },
]

job_cat_example_prompt = ChatPromptTemplate.from_messages(
    [("human", "Job Title: {title}\nDescription: {description}"), ("ai", "{category}")]
)
job_cat_few_shot_prompt = FewShotChatMessagePromptTemplate(
```

```

        example_prompt=job_cat_example_prompt,
        examples=job_category_few_shot,
    )

job_category_prompt = ChatPromptTemplate.from_messages(
    [
        ("system",
         "Given a job title and description, categorize the job into ONE of the following "
         f"domains: {'', ' '.join(JOB_CATEGORIES)}. If unsure, use 'Other'. "
         "Reply with ONLY the single category label, nothing else."),
        job_cat_few_shot_prompt,
        ("human", "Job Title: {title}\nDescription: {description}"),
    ]
)
job_category_chain = job_category_prompt | llm | StrOutputParser()

def classify_job_category(title: str, description: str) -> str:
    raw = job_category_chain.invoke(
        {"title": title, "description": str(description)[:2500]}
    ).strip()
    for cat in JOB_CATEGORIES:
        if cat.lower() in raw.lower():
            return cat
    return raw.split("\n")[0][:40] if raw else "Other"

# Expected Output: show it works for a sample datapoint.
job_sample = jobs_df.iloc[0]
sample_category = classify_job_category(job_sample["Job_Title"], job_sample["Job_Description"])
print("Job title:", job_sample["Job_Title"])
print("Predicted category:", sample_category)

```

Job title: Flutter Developer
Predicted category: Technology/IT

Step 3: Define the Requirements Extraction Task (30 marks)

A single structured prompt (JSON mode) extracts `Skills`, `Education`, and `Experience` together, with "Not specified" as the fallback when a field isn't mentioned.

```

In [12]: class JobRequirements(BaseModel):
    skills: list[str] = Field(description="Key skills, technologies, or tools mentioned.")
    education: str = Field(description="Minimum education level required/preferred, or 'Not specified'.")
    experience: str = Field(description="Years of experience or experience level required, or 'Not specified'.")

requirements_prompt = ChatPromptTemplate.from_messages(
    [
        ("system",
         "Extract the required skills, education level, and years of experience from the job "
         "description below. Respond with ONLY a JSON object of the form: "
         "'{\"skills\": [\"skill1\", \"skill2\", ...], \"education\": \"...\", \"experience\": \"...\"}'. "
         "If a field is not mentioned, use the string \"Not specified\" for it "
         "(empty list [] for skills if none found)."),
        ("human", "Job Title: {title}\nJob Description:\n{description}"),
    ]
)
requirements_chain = requirements_prompt | llm_json | StrOutputParser()

def as_text(value) -> str:
    # The model sometimes returns a list (e.g. ["Bachelor's", "Master's"]); store it as one string.
    if isinstance(value, list):
        value = ", ".join(str(v).strip() for v in value if str(v).strip())
    return str(value or "Not specified").strip() or "Not specified"

def extract_requirements(title: str, description: str) -> dict:
    # 8000 chars covers 99% of postings. Job boards often list "Education:"/"Experience:" near the
    # end, and a 3000-char cut dropped that part for 1 in 5 postings.
    raw = requirements_chain.invoke(
        {"title": title, "description": str(description)[:8000]}
    ).strip()
    try:
        data = json.loads(raw)
        skills = data.get("skills") or []
        if isinstance(skills, str): # e.g. "Python, SQL" instead of a list
            skills = skills.split(",")
        return {
            "skills": [str(s).strip() for s in skills if str(s).strip()],
            "education": as_text(data.get("education")),
        }
    
```

```

        "experience": as_text(data.get("experience")),
    }
except (json.JSONDecodeError, AttributeError):
    return {"skills": [], "education": "Not specified", "experience": "Not specified"}

# Expected Output: show it works for a sample datapoint.
sample_reqs = extract_requirements(job_sample["Job_Title"], job_sample["Job_Description"])
print(json.dumps(sample_reqs, indent=2))
{
  "skills": [
    "Flutter",
    "Software Development"
  ],
  "education": "Not specified",
  "experience": "1 year (Preferred)"
}

```

Step 4 & 5: Apply the LLM Chain to Each Job Posting and Update the DataFrame

(10 + 5 marks)

```

In [13]: JOBS_CHECKPOINT = Path("job_postings_first25_checkpoint.csv")

def run_jobs_pipeline(df: pd.DataFrame, checkpoint_path: Path, desc: str) -> pd.DataFrame:
    if checkpoint_path.exists():
        result = pd.read_csv(checkpoint_path)
    else:
        result = df.copy()
        result["Predicted_Category"] = pd.NA
        result["Required_Skills"] = pd.NA
        result["Education_Required"] = pd.NA
        result["Experience_Required"] = pd.NA

    for idx in tqdm(result.index, desc=desc):
        if pd.notna(result.loc[idx, "Predicted_Category"]):
            continue # resume support
        title = result.loc[idx, "Job_Title"]
        desc_text = result.loc[idx, "Job_Description"]

```

```

try:
    category = classify_job_category(title, desc_text)
    reqs = extract_requirements(title, desc_text)
except Exception as e:
    print(f"Row {idx} failed: {e}")
    category = "Error"
    reqs = {"skills": [], "education": "Not specified", "experience": "Not specified"}
result.loc[idx, "Predicted_Category"] = category
result.loc[idx, "Required_Skills"] = json.dumps(reqs["skills"])
result.loc[idx, "Education_Required"] = reqs["education"]
result.loc[idx, "Experience_Required"] = reqs["experience"]
result.to_csv(checkpoint_path, index=False)

result["Required_Skills"] = result["Required_Skills"].apply(json.loads)
return result

```

```

jobs_results = run_jobs_pipeline(jobs_df, JOBS_CHECKPOINT, "Part 2: first 25 postings")
jobs_results.head()

```

Out[13]:

	Job_Title	Job_Description	Predicted_Category	Required_Skills	Education_Required	Experience_Required
0	Flutter Developer	We are looking for hire experts flutter developer. So you are eligible this post then apply your resume.\nJob Types:...	Technology/IT	[Flutter, Software Development]	Not specified	1 year (Preferred)
1	Django Developer	PYTHON/DJANGO (Developer/Lead) - Job Code(PDJ - 04)\nStrong Python experience in API development (REST/RPC).\nExperi...	Technology/IT	[Python, API development, Django, Flask, Linux, SQL, PyUnit, Automated unit testing]	Not specified	Not specified
2	Machine Learning	Data Scientist (Contractor)\n\nBangalore, IN\n\nResponsibilities\n\nW e are looking for a capable data scientist to j...	Technology/IT	[Machine Learning, Deep Learning, Python, Java, Software development, Data analysis, Data collection, Algorithmic ap...	Graduate, M.Sc. in Computer Science, Mathematics or equivalent	3 years
3	iOS Developer	JOB DESCRIPTION:\n\nStrong framework outside of iOS is always a plus\n\niOS experience and generalist engineers with...	Technology/IT	[Objective-C, Cocoa Touch, Core Data, Core Animation, Core Graphics, Core Text, Networking, Mobile Network Issues, C...	Not specified	5-10 years
4	Full Stack Developer	job responsibility full stack engineer – react role make impact petsmart transforming engineering team meet need rap...	Technology/IT	[React, J Native, JavaScript, HTML, CSS, RESTful APIs, HTTP, Networking, Full Stack Web Development, MVC, Object-Ori...	Not specified	5+ years

```
In [14]: jobs_final = jobs_results[
    ["Job_Title", "Job_Description", "Predicted_Category",
     "Required_Skills", "Education_Required", "Experience_Required"]
]
jobs_final.to_csv("job_postings_analyzed_first25.csv", index=False)
print(jobs_final.shape)
jobs_final.head(10)
```

(25, 6)

Out[14]:

	Job_Title	Job_Description	Predicted_Category	Required_Skills	Education_Required	Experience_Required
0	Flutter Developer	We are looking for hire experts flutter developer. So you are eligible this post then apply your resume.\nJob Types:...	Technology/IT	[Flutter, Software Development]	Not specified	1 year (Preferred)
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3	iOS Developer	JOB DESCRIPTION:\n\nStrong framework outside of iOS is always a plus\n\niOS experience and generalist engineers with...	Technology/IT	[Objective-C, Cocoa Touch, Core Data, Core Animation, Core Graphics, Core Text, Networking, Mobile Network Issues, C...	Not specified	5-10 years
4	Full Stack Developer	job responsibility full stack engineer – react role make impact petsmart transforming engineering team meet need rap...	Technology/IT	[React, J Native, JavaScript, HTML, CSS, RESTful APIs, HTTP, Networking, Full Stack Web Development, MVC, Object-Ori...	Not specified	5+ years
5	Java Developer	Software Developer - Integration*\n\nImmediate Opening!*\n\nA dynamic Akron / Cleveland area company is looking for an e...	Technology/IT	[Proven technical expertise in the design, development, coding, testing, and debugging of enterprise software, Stron...	Bachelor's Degree in Computer Science, Information Systems, or related field, or combination of education and equiva...	2 years
6	Full Stack Developer	senior full stack developer \n-1800026h cwt looking senior full stack developer proven back-end skill well strong fr...	Technology/IT	[NodeJS, Java, NoSQL solutions, MongoDB, Elasticsearch, Redis, React, Angular, Single-page application, SPA developm...	B.Sc degree in Computer Science or Engineering	Minimum 2 years of experience
7	JavaScript Developer	Job Description:\n\nReactJS + NodeJs, Azure Functions, and GraphQL capability\n\nStrong hands on experience on javas...	Technology/IT	[ReactJS, NodeJS, Azure Functions, GraphQL, HTML5, CSS3, JavaScript]	Any graduation, Any PG, Any Doctorate	3 - 8 years
8	DevOps Engineer	Main Responsibilities and Deliverables:\nManage/support the rollout, scalability and execution of the automation as ...	Technology/IT	[Bash, Ruby, Python, Java, Puppet, Chef, Cloudify, CFEngine, Cobbler, Foreman, PHP, Linux, Windows, application debu...	Not specified	5-10 years

	Job_Title	Job_Description	Predicted_Category	Required_Skills	Education_Required	Experience_Required
9	Software Engineer	Overview\n\nBased in Silicon Valley, Tintri is a wholly owned subsidiary of DataDirect Networks (DDN.com), the dat...	Technology/IT	[REST API, C/C++, Python, Go, Git, Gerrit, Jenkins]	BS or MS; computer engineering, computer science or related technical field	7 years

Verify the outputs (spot-check)

Compare a few postings' extracted fields with their description text, then check coverage across all 25 postings.

```
In [15]: spot = jobs_final.sample(3, random_state=42)
for _, row in spot.iterrows():
    print(f"=== {row['Job_Title']} -> {row['Predicted_Category']}")
    print("Description:", " ".join(str(row["Job_Description"]).split()[:400], "...")
    print("Skills      :", row["Required_Skills"])
    print("Education   :", row["Education_Required"])
    print("Experience  :", row["Experience_Required"])
    print()

n = len(jobs_final)
print(f"Postings with at least one skill: {(jobs_final['Required_Skills'].apply(len) > 0).sum()}/{n}")
print(f"Education specified           : {(jobs_final['Education_Required'] != 'Not specified').sum()}/{n}")
print(f"Experience specified            : {(jobs_final['Experience_Required'] != 'Not specified').sum()}/{n}")
```

=== DevOps Engineer -> Technology/IT

Description: Main Responsibilities and Deliverables: Manage/support the rollout, scalability and execution of the automation as it is consumed by the various teams Ensure automation services scales to handle rapid growth Using the automations, establish the ability to measure and optimize system health and performance. On an ongoing basis; configure, tune and troubleshoot automations to achieve optimal applica ...

Skills : ['Bash', 'Ruby', 'Python', 'Java', 'Puppet', 'Chef', 'Cloudify', 'CFEngine', 'Cobbler', 'Foreman', 'PHP', 'Linux', 'Windows', 'application debugging', 'performance', 'scalability', 'capacity planning', 'Infrastructure as Code', 'Urban Code', 'Hadoop', 'Tomcat', 'Mule', 'OpenAM', 'Apache', 'F5 load balancers', 'networking theory', 'TCP/IP', 'HTTP', 'NTP', 'DNS']

Education : Not specified

Experience : 5-10 years

=== Wordpress Developer -> Technology/IT

Description: Experience: 2-5 years Job Location:- Aurangabad/Pune Vacancies:- 02 Note: Fresher Do Not Apply Job Description Looking for experienced developers who are passionate to work with an IT / Software Development company. Basic Requirements: Having prior working experience on WordPress Should be proficient verbally and written communication skills. Should be capable of writing an efficient code using be ...

Skills : ['WordPress', 'Plugin-in development', 'PHP', 'HTML/HTML5', 'Javascript/jQuery', 'Bootstrap', 'MySQL']

Education : UG: B.Sc (CS/CSC/IT), BCA, BCS, BE, B.Tech (CS/CSE/IT), M.Sc (CS/CSC/IT), MCA, MCS, ME, M.Tech (CS/CSE/IT)

Experience : 2-5 years

=== Flutter Developer -> Technology/IT

Description: We are looking for hire experts flutter developer. So you are eligible this post then apply your resume. Job Types: Full-time, Part-time Salary: ₹20,000.00 - ₹40,000.00 per month Benefits: Flexible schedule Food allowance Schedule: Day shift Supplemental Pay: Joining bonus Overtime pay Experience: total work: 1 year (Preferred) Housing rent subsidy: Yes Industry: Software Development Work Remotely ...

Skills : ['Flutter', 'Software Development']

Education : Not specified

Experience : 1 year (Preferred)

Postings with at least one skill: 25/25

Education specified : 15/25

Experience specified : 22/25

Bonus: Full-Dataset Runs (up to 20 marks)

Runs the same pipelines over **all** rows of both datasets (2,225 BBC articles + 2,277 job postings), using local Ollama so there's no rate limiting. Each pipeline checkpoints to CSV after every row, so it can be safely interrupted and resumed by re-running the cell.

⚠ This is the slow part — expect this to take a while on CPU. Run it last / let it run in the background.

```
In [16]: BBC_FULL_CHECKPOINT = Path("bbc_news_full_checkpoint.csv")

bbc_full_df = bbc_df_full.reset_index(drop=True)
bbc_full_df.insert(0, "Article_ID", bbc_full_df.index)

bbc_full_results = run_bbc_pipeline(bbc_full_df, BBC_FULL_CHECKPOINT, "BONUS Part 1: all articles")
bbc_full_final = bbc_full_results[
    ["Article_ID", "category", "filename", "title", "content", "Detected_Topic", "Summary", "Key_Entities"]
]
bbc_full_final.to_csv("bbc_news_analyzed_full.csv", index=False)
print(bbc_full_final.shape)
```

(2225, 8)

```
In [17]: JOBS_FULL_CHECKPOINT = Path("job_postings_full_checkpoint.csv")

jobs_full_df = jobs_df_full.reset_index(drop=True)

jobs_full_results = run_jobs_pipeline(jobs_full_df, JOBS_FULL_CHECKPOINT, "BONUS Part 2: all postings")
jobs_full_final = jobs_full_results[
    ["Job_Title", "Job_Description", "Predicted_Category",
     "Required_Skills", "Education_Required", "Experience_Required"]
]
jobs_full_final.to_csv("job_postings_analyzed_full.csv", index=False)
print(jobs_full_final.shape)
```

(2277, 6)